Two mere propositions that imply each other are equivalent

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 294-296 and its proof environment opens on line 297, the next line; the pairing is by adjacency and the retained proof range 297-301 is the whole environment.

Editorial note. The retained counter values (chapter 3, section 3, theorem offset 2) make the counted lemma print as Lemma 3.3.3, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel lem:equiv-iff-hprop = 3.3.3).

Editorial note. No context span and no external reference are needed: the retained statement and proof use no label from outside their own ranges.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: mutual implication between mere propositions is equivalence (author label lem:equiv-iff-hprop, book Lemma 3.3.3).

Allowed standard prerequisites (editorial).

- Mere proposition. A type is a mere proposition when any two of its elements are equal.. It is the hypothesis on both sides.
- Quasi-inverse criterion. A map is an equivalence as soon as it has a two-sided inverse.. It is how the equivalence is produced from the two implications.

Lemma 3.3.3. *If P and Q are mere propositions such that $P \to Q$ and $Q \to P$, then $P \simeq Q$.*

Proof. Suppose given $f : P \to Q$ and $g : Q \to P$. Then for any $x : P$, we have $g(f(x)) = x$ since P is a mere proposition. Similarly, for any $y : Q$ we have $f(g(y)) = y$ since Q is a mere proposition; thus f and g are quasi-inverses. □